

Assignment 2

ME Analysis 1, Fall 2026

Return in class on Oct. 5.

100 points

- 1) Show that, $f_1(x) = e^x$ and $f_2(x) = \sin x$ are orthogonal to each other in the interval, $[\pi/4, 5\pi/4]$.
- 2) Prove orthogonality of the set of functions, $\{\cos(2n+1)x\}$, in the interval, $[0, \pi/2]$. $n = 0, 1, 2, \dots$. Obtain the norm of these functions.
- 3) $\{\phi_n(x)\}$, $n = 0, 1, 2, \dots$, is a set of orthogonal functions on the interval, $[a, b]$. Prove that $\|\phi_m(x) + \phi_n(x)\|^2 = \|\phi_m(x)\|^2 + \|\phi_n(x)\|^2$, for $m \neq n$.
- 4) Find the real Fourier series for:

$$f(x) = \begin{cases} 0, & \text{if } -\pi < x < 0 \\ x^2, & \text{if } 0 \leq x < \pi \end{cases}$$

Using this result, show that, $\pi^2/12 = 1 - 1/2^2 + 1/3^2 - 1/4^2 + \dots$. Hint: Utilize the value of function at $x = 0$ in the series.

- 5) Find the complex Fourier series for:

$$f(x) = \begin{cases} e^x, & \text{if } -1 < x < 0 \\ e^{-x}, & \text{if } 0 < x < 1 \end{cases}$$

Here, $f(x) = e^{-|x|}$. Plot the spectrum at the resulting component frequencies.

- 6) Vector algebra:

- Vector, $\overrightarrow{P_1P_2} = 4\hat{i} + 8\hat{j}$, has its initial point at $(-3, 10)$. Find its terminal point.
- Vector, $\overrightarrow{P_1P_2} = -5\hat{i} - \hat{j}$, has its terminal point at $(4, 7)$. Find its initial point.

- 7) Vector algebra:

- Vector $\vec{a} = 3\hat{i} + 7\hat{j}$. Find another vector, $\vec{b}$, which is parallel to $\vec{a}$, and has magnitude, 2.
- Show that $\vec{a} + \vec{b} + \vec{c} + \vec{d} = 0$, in Fig. 1. Hint: Utilize the diagonal of the quadrilateral.

- 8) Vector algebra:

- Prove that the diagonals of the rhombus in Fig. 2 are orthogonal to each other. Hint: Adjacent edges of rhombus have equal length.
- Find $\vec{V} = x_1\hat{i} + y_1\hat{j} + \hat{k}$ such that it is perpendicular to both $\vec{a} = \langle 3, 1, -1 \rangle$ and $\vec{b} = \langle -3, 2, 2 \rangle$.
- Are the points $P_1(1, 1, -2)$, $P_2(4, 0, -3)$, $P_3(1, -5, 10)$, and $P_4(-7, 2, 4)$ coplanar? Explain your answer.

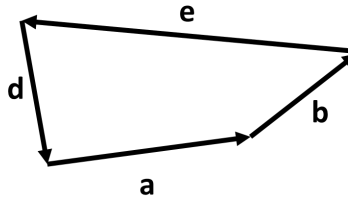


Fig. 1 Vector sum problem

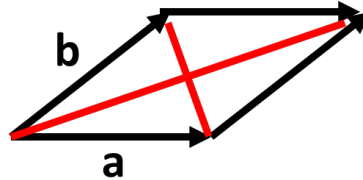


Fig. 2 Rhombus diagonal problem

9) Lines:

- Find parametric equations for the line through $(6, 4, -2)$ that is parallel to the line $x/2 = (1 - y)/3 = (z - 5)/6$.
- Find the angle between the two lines: $x = 4 - t, y = 3 + 2t, z = -2t$ and $x = 5 + 2s, y = 1 + 3s, z = 5 - 6s$.

10) Planes:

- Find equation of the plane containing the lines: $x = 1 + 3t, y = 1 - t, z = 2 + t$ and $x = 4 + 4s, y = 2s, z = 3 + s$.
Hint: Obtain the normal to the plane from direction vectors of lines, and use point-normal form.
- Find equation of the plane containing the line, $x = 4 + 3t, y = -t, z = 1 + 5t$, and is orthogonal to the plane, $x + y + z = 7$. Hint: Normal to the plane will be orthogonal to the line and the normal to the given plane.